Sliding Window Maximum (hard)

Problem Statement

You are given an array of integers `nums`, and integer `k`, size of the sliding window which moves from the very `left` to the very `right` in the array. In each window, you can have only `k` numbers, and the window moves one position right by each time.

Return the array containing the `maximum element` of each sliding window.

Examples

Example 1:

- **Input**: nums = [2, 1, 5, 1, 3, 2], k = 3
- **Expected Output**: [5, 5, 5, 3]
- **Justification**: Here, the window of size 3 moves across the array. The maximum values in each window are 5 (from [2, 1, 5]), 5 (from [1, 5, 1]), 5 (from [5, 1, 3]), and 3 (from [1, 3, 2]).

Example 2:

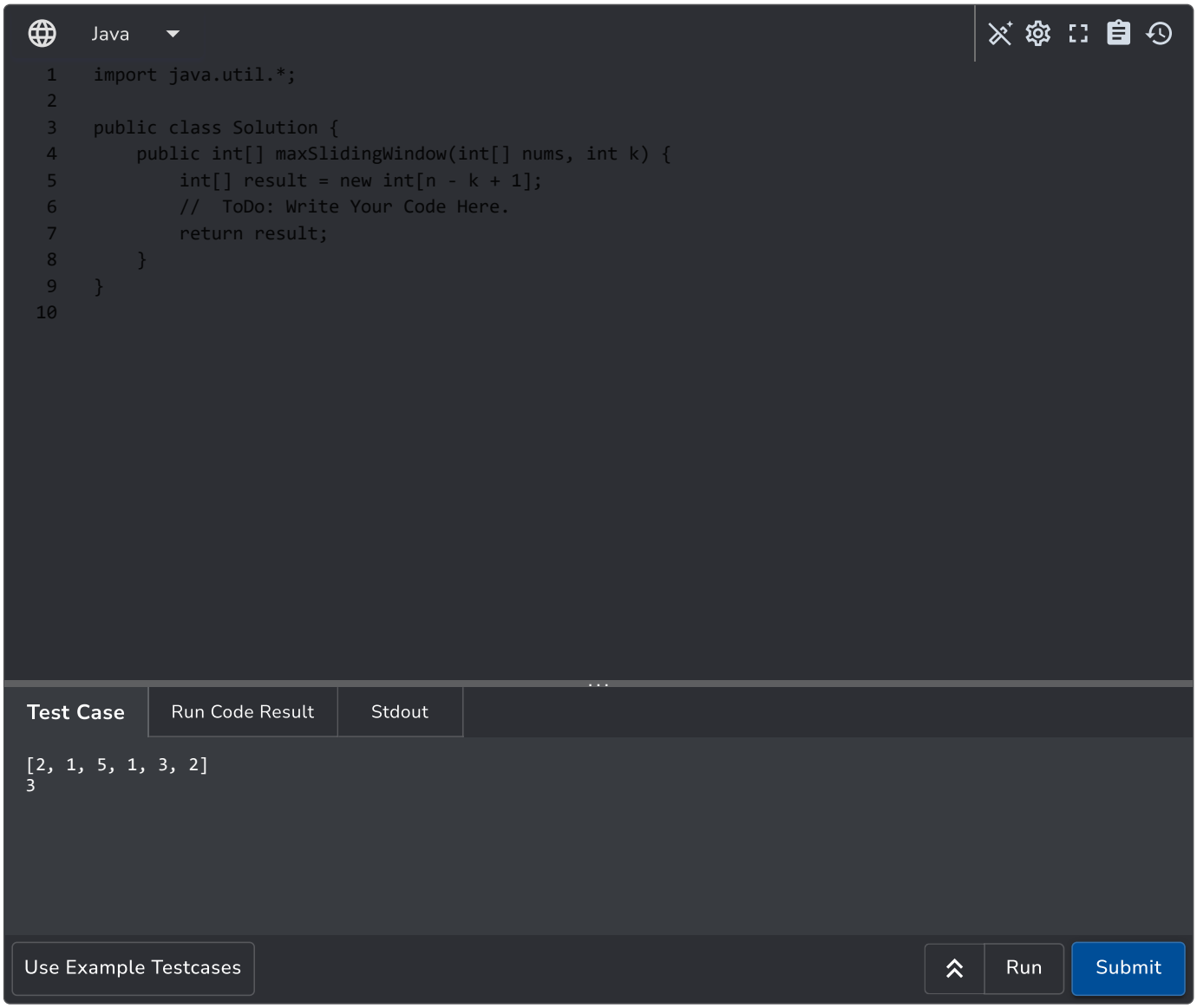
- **Input**: nums = [4, -2, -3, 4, 1], k = 2
- **Expected Output**: [4, -2, 4, 4]
- **Justification**: The maximum values in each window of size 2 are 4 (from [4, -2]), -2 (from [-2, -3]), 4 (from [-3, 4]), and 4 (from [4, 1]).

Example 3:

- **Input**: nums = [9, 7, 2, 4, 6], k = 4
- **Expected Output**: [9, 7]
- **Justification**: In this case, there are only two windows of size 4. The maximum value in the first window is 9 (from [9, 7, 2, 4]), and in the second window is 7 (from [7, 2, 4, 6]).

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Solution: Sliding Window Maximum

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